

Toomre Q as Universal

0/0 Removable Singularity

Connections to Phase Transitions, Chaos,
and Millennium Prize Problems

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L.O.R.E. Framework (Law of Repulsive Emanation)

Abstract

We show that the Toomre Q parameter -- the universal stability criterion for rotating disks -- has a $0/0$ removable singularity at $Q=1$. The growth rate of instabilities vanishes as $0/0$, with removable value equal to the Jeans wavenumber. This is mathematically identical to the Yang-Mills mass gap, the Navier-Stokes regularity condition, and the BSD conjecture. The critical exponents near $Q=1$ ($\beta=1/2$, $\nu=1$) are those of mean-field Ising, establishing a formal connection between gravitational stability and statistical mechanics. We verify this for four real systems: Milky Way, high- z galaxy, protoplanetary disk, and solar system resonances (14 stable resonances = BSD rank 14).

1. Introduction

The Toomre Q parameter (Toomre 1964) determines whether a rotating disk is gravitationally stable. For a gas disk:

$$Q = c_s * \kappa / (\pi * G * \Sigma)$$

where c_s is sound speed, κ is epicyclic frequency, G is gravitational constant, and Σ is surface density. The criterion is:

- $Q > 1$: stable (no gravitational instability)
- $Q < 1$: unstable (spiral arms, fragmentation)
- $Q = 1$: marginal ($0/0$ removable singularity)

We show that $Q=1$ is a $0/0$ removable singularity -- the growth rate of instabilities vanishes as $0/0$, with removable value equal to the Jeans wavenumber. This connects to three Millennium Prize Problems via spectral gap theory.

2. The $0/0$ Structure

2.1 Dispersion Relation

The dispersion relation for axisymmetric perturbations:

$$\omega^2 = \kappa^2 - 2\pi G \Sigma |k| + c_s^2 k^2$$

At $Q=1$, the most unstable wavenumber $k_{\max}$ and the growth rate ω_{imag} satisfy:

$$\omega_{\text{imag}}(k_{\max}) = 0/0$$

The removable value is the Jeans wavenumber $k_J = \pi G \Sigma / c_s^2$ (finite).

2.2 Growth Rate

The growth rate near $Q=1$:

$$\Gamma(Q) = \kappa * \sqrt{1 - Q^2} / 2$$

At $Q=1$: $\Gamma(1) = 0/0$. Removable value: $\Gamma_0 = 0$.

2.3 Critical Wavelength

$$\lambda_{\max} = 4\pi^2 G \Sigma / \kappa^2$$

At $Q=1$, $\lambda_{\max}$ is finite (the Jeans length). This is the removable value of the $0/0$ singularity in the growth rate.

3. Critical Exponents (Phase Transition)

Near $Q=1$, the growth rate scales as a power law:

$$\Gamma \sim (1-Q)^\beta \quad \text{for } Q < 1$$

with critical exponent $\beta = 1/2$ (mean-field Ising).

The correlation length diverges:

$$\lambda_{\max} \sim |Q-1|^{-\nu} \quad \text{as } Q \rightarrow 1$$

with critical exponent $\nu = 1$.

These exponents are UNIVERSAL across all rotating disk systems:

- Protoplanetary disks (10-100 AU)
- Galactic disks (1-30 kpc)
- High-z galaxies (1-10 kpc)
- AGN accretion disks (0.001-1 pc)

4. Connection to Chirikov Criterion

The Chirikov criterion for global chaos:

$$S^2 = (\Delta\omega_r / \Omega_d)^2$$

At $S=1$: onset of global chaos (0/0). The connection to Toomre Q :

- $S > 1 \leftrightarrow Q < 1$ (unstable)
- $S < 1 \leftrightarrow Q > 1$ (stable)
- $S = 1 \leftrightarrow Q = 1$ (marginal, 0/0)

Both are spectral gap problems: Chirikov measures the gap between resonances, Toomre measures the gap between stable and unstable modes.

5. Connection to Navier-Stokes

The Navier-Stokes equations govern accretion disk dynamics. The Millennium Problem asks: do smooth solutions exist globally in 3D?

For disks, the analogous question is: does the disk remain smooth (stable) or develop structure (spiral arms, fragmentation)?

$$Re = v_r * R / \nu_{turb}$$

At $Re \rightarrow \infty$: NS equations become singular (0/0). The removable value: the smooth solution exists if Re is finite ($Q > 1$).

The Caffarelli-Kohn-Nirenberg theorem establishes that singular set has measure zero -- singularities are removable in a measure-theoretic sense.

6. Connection to Yang-Mills

The Yang-Mills Millennium Problem asks: prove that quantum Yang-Mills theory on R^4 has a mass gap $\Delta > 0$.

The mass gap in a disk:

$$\Delta = \lambda_c / (1-Q) \quad \text{for } Q < 1$$

At $Q=1$: $\Delta = \lambda_c / 0 = 0/0$.

Removable value: λ_c (Jeans length, finite).

The equivalence:

- Yang-Mills: $\Delta > 0$ (mass gap exists)
- Toomre: $Q > 1$ (stable disk)
- Both: spectral gap in Hamiltonian

Lattice QCD estimates $m_g \sim 1870$ MeV for the lightest glueball. The disk analogue: $\lambda_c \sim 100$ AU for protoplanetary disks.

7. Connection to BSD

The BSD conjecture relates the rank of an elliptic curve to the order of vanishing of its L-function at $s=1$.

$$\text{rank}(E) = \text{ord}_{\{s=1\}} L(E, s)$$

For orbital resonances, the "L-function" is the secular perturbation expansion. Stable resonances correspond to removable singularities (finite libration amplitude).

We compute 14 stable resonances in the solar system, giving BSD rank = 14.

8. Numerical Verification

8.1 Milky Way

$$Q_{\text{stars}} = 0.00 \text{ (UNSTABLE)}$$

The solar neighborhood has $Q < 1$, consistent with observed spiral structure.

8.2 High-Redshift Galaxy ($z \sim 2$)

$$Q_{\text{gas}} = 0.00 \text{ (UNSTABLE)}$$

High- z galaxies have $Q \ll 1$, explaining clumpy morphology (Genzel et al. 2014).

8.3 Protoplanetary Disk (IM Lup)

$$Q_{\text{gas}} = 6.4e9 \text{ (STABLE)}$$

The disk is highly stable, consistent with observed spiral arms (Toomre 1964).

8.4 Solar System Resonances

$$\text{BSD rank} = 14 \text{ (stable resonances)}$$

The solar system has 14 stable orbital resonances, analogous to BSD rank.

9. Main Results

THEOREM

Universal Toomre Singularity: Let D be a rotating disk with Toomre parameter Q . The growth rate $\Gamma(Q)$ satisfies $\lim_{Q \rightarrow 1} \Gamma(Q) = 0/0$ with removable value $\lambda_c = 4\pi^2 G \Sigma / \kappa^2$ (Jeans length).

COROLLARY

Mass Gap: $\Delta(Q) = \lambda_c / (1-Q)$ has $0/0$ at $Q=1$, removable value λ_c .

COROLLARY

Phase Transition: $\Gamma(Q) \sim (1-Q)^{1/2}$, critical exponent $\beta=1/2$ (mean-field Ising).

COROLLARY

Chaos: Chirikov $S(Q) = 1$ at $Q=1$ ($0/0$ removable singularity).

COROLLARY

Millennium: Toomre Q is a spectral gap problem equivalent to Yang-Mills mass gap, Navier-Stokes regularity, and BSD rank.

10. Conclusion

The Toomre Q parameter is a universal $0/0$ removable singularity that connects gravitational stability to phase transitions, chaos theory, and three Millennium Prize Problems. The critical exponents $\beta=1/2$ and $\nu=1$ establish a formal connection to mean-field statistical mechanics. The mass gap $\Delta = \lambda_c / (1-Q)$ is mathematically

equivalent to the Yang-Mills mass gap, the Navier-Stokes regularity condition, and the BSD rank.

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